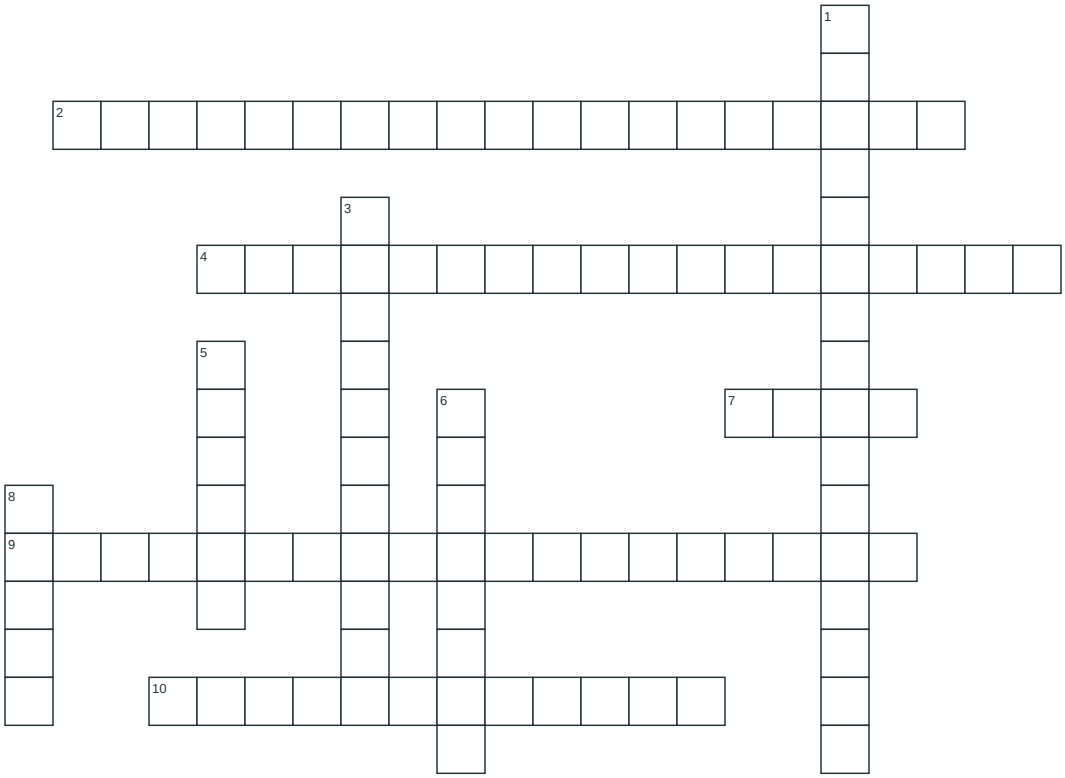
Modern Physics Fundamentals Crossword

Easy | 10 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

2. Quantum objects can display both wave-like and particle-like behavior
4. Predicts likelihoods of measurement outcomes
7. Measure of the amount of matter in an object.
9. Emission of electrons from a material after light transfers sufficient energy
10. The directed change from initial to final position

DOWN

1. Specific wavelengths of light emitted by excited atoms
3. An electron can occupy certain allowed energies
5. Quantum, or discrete packet, of electromagnetic energy
6. The total length of a traveled path
8. Quantity that is a scalar

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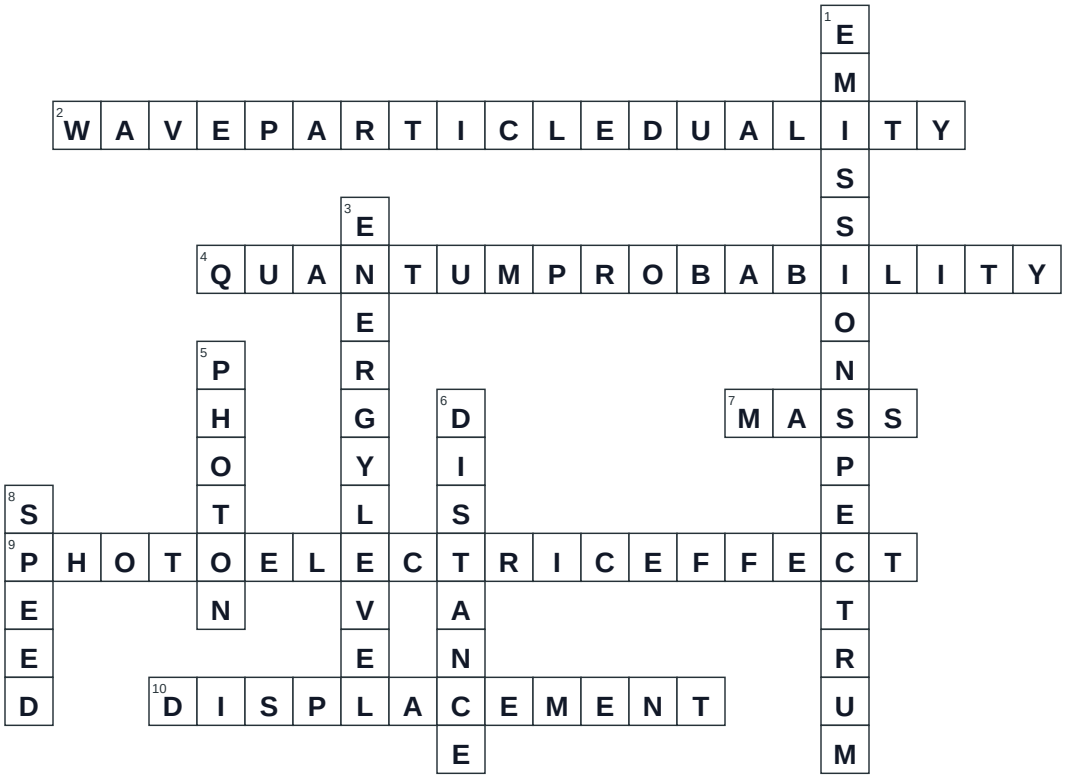
Modern Physics Fundamentals Crossword

Easy | 10 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

2. Wave-particle Duality
4. Quantum Probability
7. Mass
9. Photoelectric Effect
10. Displacement

DOWN

1. Emission Spectrum
3. Energy Level
5. Photon
6. Distance
8. Speed

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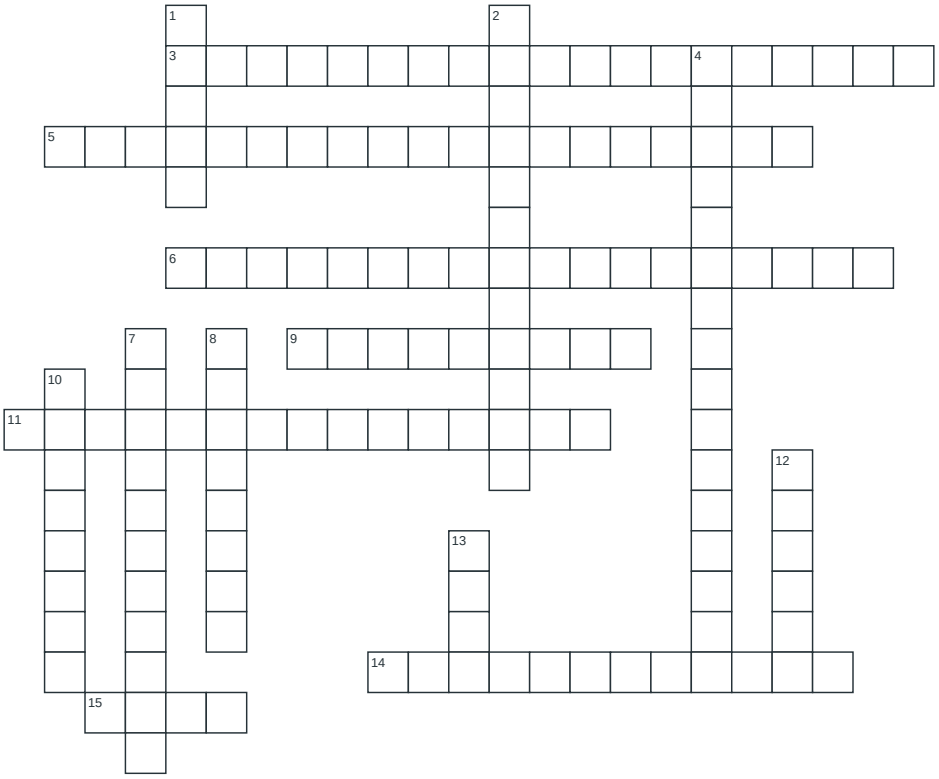
Modern Physics Fundamentals Crossword

Medium | 15 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

3. Emission of electrons from a material after light transfers sufficient energy
5. Quantum objects can display both wave-like and particle-like behavior
6. Predicts likelihoods of measurement outcomes
9. Orientation of motion, needed to describe a velocity vector.
11. SI-derived unit commonly used to measure speed or velocity.
14. The directed change from initial to final position
15. State in which position does not change relative to a chosen reference point.

DOWN

1. Quantity that is a scalar
2. Change in velocity per unit time.
4. Specific wavelengths of light emitted by excited atoms
7. An electron can occupy certain allowed energies
8. The total length of a traveled path
10. The vector sum of all forces acting on an object
12. Quantum, or discrete packet, of electromagnetic energy
13. Measure of the amount of matter in an object.

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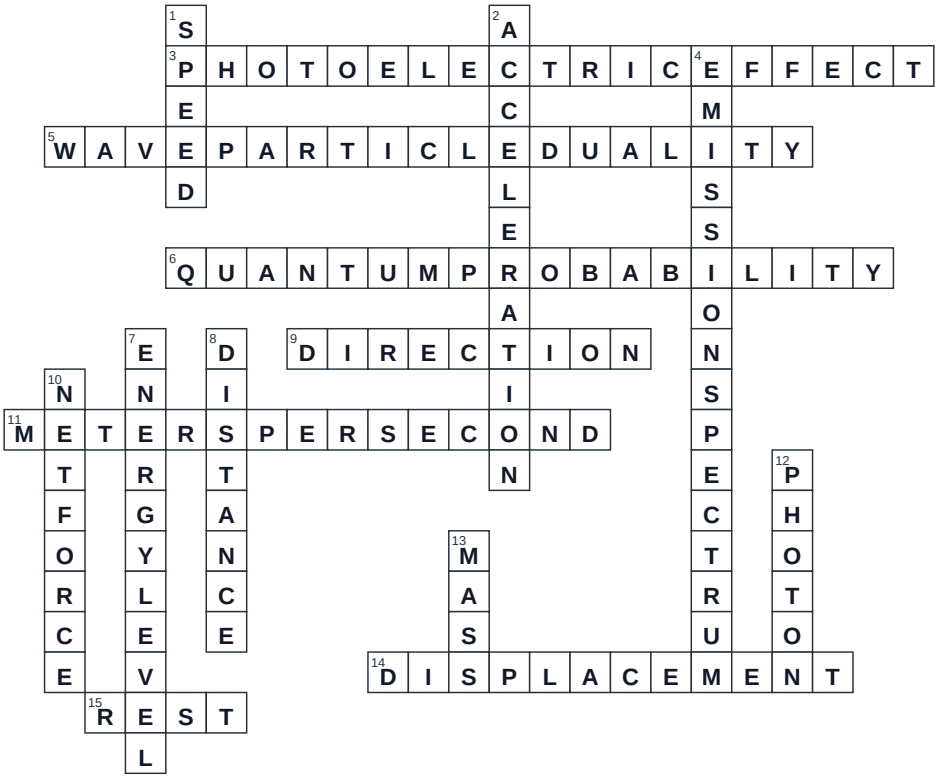
Modern Physics Fundamentals Crossword

Medium | 15 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

3. Photoelectric Effect
5. Wave-particle Duality
6. Quantum Probability
9. Direction
11. Meters per second
14. Displacement
15. Rest

DOWN

1. Speed
2. Acceleration
4. Emission Spectrum
7. Energy Level
8. Distance
10. Net Force
12. Photon
13. Mass

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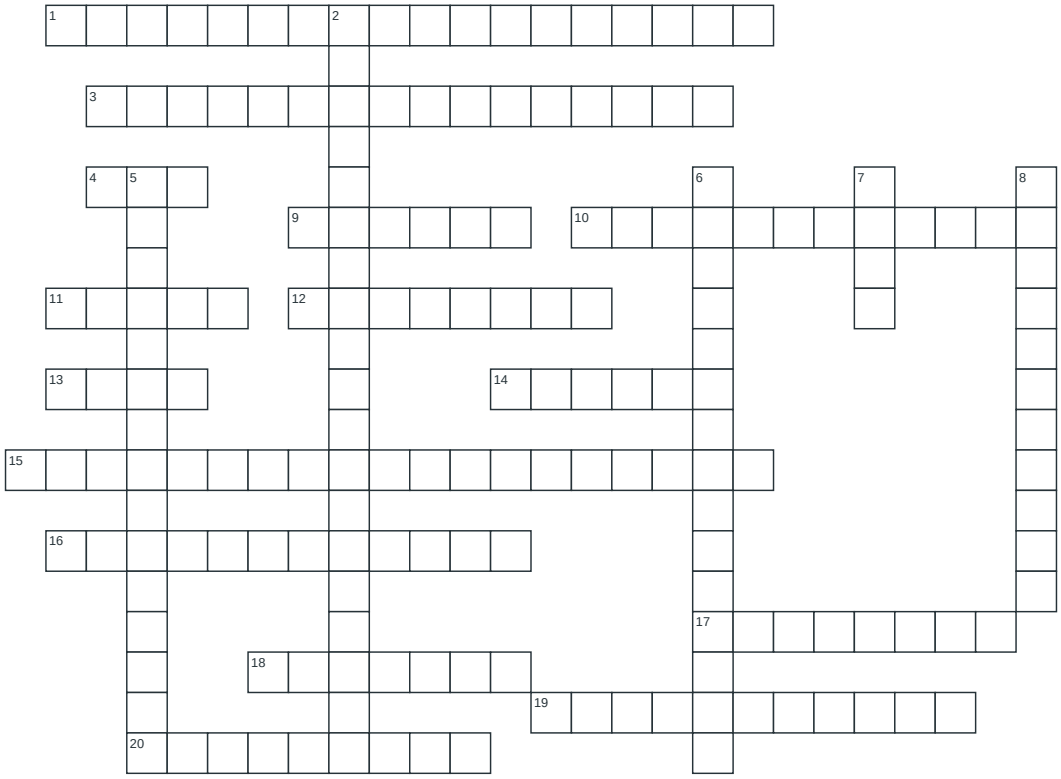
Modern Physics Fundamentals Crossword

Hard | 20 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

1. Predicts likelihoods of measurement outcomes
3. Specific wavelengths of light emitted by excited atoms
4. Relationship that expresses Newton's second law for constant mass
9. The gravitational force acting on an object
10. Change in velocity per unit time.
11. Quantity that is a scalar
12. The vector sum of all forces acting on an object
13. State in which position does not change relative to a chosen reference point.
14. Quantum, or discrete packet, of electromagnetic energy
15. Quantum objects can display both wave-like and particle-like behavior
16. The directed change from initial to final position

ACROSS (continued)

17. The total length of a traveled path
18. Resistance to changes in motion
19. Force that is a surface's push perpendicular to that surface
20. Orientation of motion, needed to describe a velocity vector.

DOWN

2. Emission of electrons from a material after light transfers sufficient energy
5. SI-derived unit commonly used to measure speed or velocity.
6. Forces between interacting objects are equal in magnitude and opposite in direction.
7. Measure of the amount of matter in an object.
8. An electron can occupy certain allowed energies

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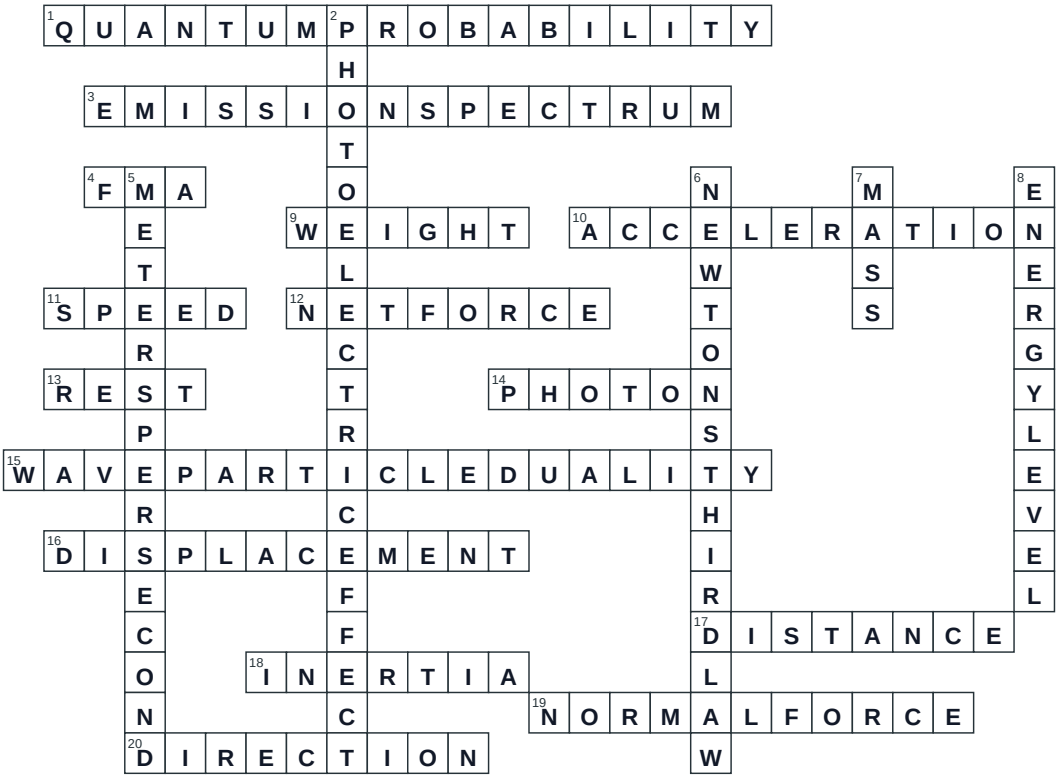
Modern Physics Fundamentals Crossword

Hard | 20 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

1. Quantum Probability
3. Emission Spectrum
4. F = ma
9. Weight
10. Acceleration
11. Speed
12. Net Force
13. Rest
14. Photon
15. Wave-particle Duality
16. Displacement

ACROSS (continued)

17. Distance
18. Inertia
19. Normal Force
20. Direction

DOWN

2. Photoelectric Effect
5. Meters per second
6. Newton’s third law
7. Mass
8. Energy Level

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